

Retarded Potentials and Far-Field Radiation

Source: *Cambridge Electromagnetism, Chapter 7 exercises, Question 4; accompanies David Tong's Electromagnetism.*

Let ϕ be the retarded scalar potential given by

$$\phi(t, \mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3y \frac{\rho(t_{\text{ret}}, \mathbf{y})}{R}, \quad (1.1)$$

where $R = |\mathbf{x} - \mathbf{y}|$, $t_{\text{ret}} = t - R/c$, and $\hat{\mathbf{R}} = (\mathbf{x} - \mathbf{y})/R$. Show that

$$\frac{\partial}{\partial t} \phi(t, \mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3y \frac{\dot{\rho}(t_{\text{ret}}, \mathbf{y})}{R}, \quad (1.2)$$

where $\dot{\rho}(t_{\text{ret}}, \mathbf{y})$ is $\partial\rho(t, \mathbf{y})/\partial t$ evaluated at t_{ret} . Show further that

$$\nabla\phi(t, \mathbf{x}) = -\frac{1}{4\pi\epsilon_0} \int d^3y \hat{\mathbf{R}} \left(\frac{1}{R^2} \rho(t_{\text{ret}}, \mathbf{y}) + \frac{1}{cR} \dot{\rho}(t_{\text{ret}}, \mathbf{y}) \right). \quad (1.3)$$

Hence verify, using $\nabla^2(1/R) = -4\pi\delta^{(3)}(\mathbf{x} - \mathbf{y})$, that ϕ satisfies the wave equation:

$$\square \phi(t, \mathbf{x}) = -\frac{1}{\epsilon_0} \rho(t, \mathbf{x}). \quad (1.4)$$

Write down a similar retarded solution for the vector potential $\mathbf{A}$ in terms of the current density $\mathbf{J}$.

Now assume that ρ and $\mathbf{J}$ are non-zero only in a finite region. Setting $\hat{\mathbf{x}} = \mathbf{x}/|\mathbf{x}|$, show that the leading terms in the far-field expansion are

$$\begin{aligned} \mathbf{E}(t, \mathbf{x}) &\approx \frac{\mu_0}{4\pi|\mathbf{x}|} \int d^3y \left(\hat{\mathbf{x}} c \dot{\rho}(t_{\text{ret}}, \mathbf{y}) - \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right) \\ &= \frac{\mu_0}{4\pi|\mathbf{x}|} \hat{\mathbf{x}} \times \left(\hat{\mathbf{x}} \times \int d^3y \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right), \end{aligned} \quad (1.5)$$

where conservation of current, integration by parts, and discarding of a surface integral have all been used, and

$$\mathbf{B}(t, \mathbf{x}) \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|} \hat{\mathbf{x}} \times \int d^3y \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) = \frac{1}{c} \hat{\mathbf{x}} \times \mathbf{E}(t, \mathbf{x}). \quad (1.6)$$

(Note that these results do not assume the dipole approximation.) Determine the Poynting vector.

Hint: when using current conservation, and integration by parts, be careful with the $\mathbf{y}$ -dependence of t_{ret} .

Solution. Following from the retarded potential, we have, trivially,

$$\frac{\partial}{\partial t} \phi(t, \mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3y \frac{\dot{\rho}(t_{\text{ret}}, \mathbf{y})}{R}. \quad (1.7)$$

Then, note that

$$\nabla \left(\frac{1}{R} \right) = -\frac{\hat{\mathbf{R}}}{R^2} \quad \text{and} \quad \nabla \rho(t_{\text{ret}}, \mathbf{y}) = \frac{\partial \rho(t_{\text{ret}}, \mathbf{y})}{\partial R} = -\frac{1}{c} \dot{\rho}(t_{\text{ret}}, \mathbf{y}) \hat{\mathbf{R}}. \quad (1.8)$$

Thus, we have

$$\nabla \phi(t, \mathbf{x}) = -\frac{1}{4\pi\epsilon_0} \int d^3y \hat{\mathbf{R}} \left(\frac{1}{R^2} \rho(t_{\text{ret}}, \mathbf{y}) + \frac{1}{cR} \dot{\rho}(t_{\text{ret}}, \mathbf{y}) \right). \quad (1.9)$$

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Now, first consider the second-order time derivative, which is simply given by

$$\frac{\partial^2}{\partial t^2} \phi(t, \mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3y \frac{\ddot{\rho}(t_{\text{ret}}, \mathbf{y})}{R}. \quad (1.10)$$

For the Laplacian, it is slightly trickier; we find

$$\nabla^2 \phi(t, \mathbf{x}) = -\frac{1}{\epsilon_0} \rho(t, \mathbf{x}) + \frac{1}{4\pi\epsilon_0 c^2} \int d^3y \frac{\ddot{\rho}(t_{\text{ret}}, \mathbf{y})}{R}. \quad (1.11)$$

Putting them together, we obtain

$$\square \phi(t, \mathbf{x}) = -\frac{1}{\epsilon_0} \rho(t, \mathbf{x}). \quad (1.12)$$

By simple analogy, we have

$$\square \mathbf{A}(t, \mathbf{x}) = -\mu_0 \mathbf{J}(t, \mathbf{x}), \quad (1.13)$$

$$\frac{\partial}{\partial t} \mathbf{A}(t, \mathbf{x}) = \frac{\mu_0}{4\pi} \int d^3y \frac{\dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y})}{R}, \quad (1.14)$$

and

$$\nabla A_i(t, \mathbf{x}) = -\frac{\mu_0}{4\pi} \int d^3y \hat{\mathbf{R}} \left(\frac{1}{R^2} J_i(t_{\text{ret}}, \mathbf{y}) + \frac{1}{cR} \dot{J}_i(t_{\text{ret}}, \mathbf{y}) \right) \quad (1.15)$$

In the far-field, we have $R \gg 1$ and $|\mathbf{x}| \approx R$, which gives

$$\begin{aligned}
\mathbf{E}(t, \mathbf{x}) &= -\nabla\phi(t, \mathbf{x}) - \frac{\partial\mathbf{A}(t, \mathbf{x})}{\partial t} \\
&\approx \frac{1}{4\pi\epsilon_0} \int d^3y \hat{\mathbf{x}} \frac{1}{c|\mathbf{x}|} \dot{\rho}(t_{\text{ret}}, \mathbf{y}) - \frac{\mu_0}{4\pi} \int d^3y \hat{\mathbf{x}} \frac{1}{|\mathbf{x}|} \dot{\mathbf{J}}_i(t_{\text{ret}}, \mathbf{y}) \\
&= \frac{\mu_0}{4\pi|\mathbf{x}|} \int d^3y \left(\hat{\mathbf{x}} c \dot{\rho}(t_{\text{ret}}, \mathbf{y}) - \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right).
\end{aligned} \tag{1.16}$$

To simplify the first term, consider the continuity equation,

$$\dot{\rho}(t_{\text{ret}}, \mathbf{y}) + \nabla_y \cdot \mathbf{J}(t, \mathbf{y}) \Big|_{t=t_{\text{ret}}} = 0. \tag{1.17}$$

However, to perform an integration by parts, we need

$$\nabla_y \cdot \mathbf{J}(t_{\text{ret}}, \mathbf{y}) = \nabla_y \cdot \mathbf{J}(t, \mathbf{y}) \Big|_{t=t_{\text{ret}}} + \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \cdot \nabla t_{\text{ret}} = \nabla_y \cdot \mathbf{J}(t, \mathbf{y}) \Big|_{t=t_{\text{ret}}} + \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \cdot \frac{\hat{\mathbf{R}}}{c}. \tag{1.18}$$

Thus, the first term integral becomes

$$\frac{\mu_0}{4\pi|\mathbf{x}|} \int d^3y \hat{\mathbf{x}} \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \cdot \hat{\mathbf{R}} - \frac{\mu_0}{4\pi|\mathbf{x}|} \oint d^2y \mathbf{J}(t_{\text{ret}}, \mathbf{y}) \cdot \hat{\mathbf{n}}. \tag{1.19}$$

Since $\mathbf{J}$ is bounded, the surface integral vanishes. Therefore, we can write

$$E(t, \mathbf{x}) \approx \frac{\mu_0}{4\pi|\mathbf{x}|} \int d^3y \left(\hat{\mathbf{x}} (\dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \cdot \hat{\mathbf{R}}) - \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right), \tag{1.20}$$

Again, for far-field, we have $\hat{\mathbf{R}} \approx \hat{\mathbf{x}}$. Finally, using the vector identity, we find

$$\mathbf{E}(t, \mathbf{x}) \approx \frac{\mu_0}{4\pi|\mathbf{x}|} \hat{\mathbf{x}} \times \left(\hat{\mathbf{x}} \times \int d^3y \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right). \tag{1.21}$$

For the magnetic field, we have

$$B_i(t, \mathbf{x}) = \epsilon_{ijk} \partial_j A_k \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|} \epsilon_{ijk} \int d^3y \hat{\mathbf{x}}_j \dot{J}_k(t, \mathbf{x}), \tag{1.22}$$

where we have made a far-field approximation again. Therefore, we find

$$\mathbf{B}(t, \mathbf{x}) \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|} \hat{\mathbf{x}} \times \int d^3y \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) = \frac{1}{c} \hat{\mathbf{x}} \times \mathbf{E}(t, \mathbf{x}). \tag{1.23}$$

The Poynting vector of the radiation is, therefore, given by

$$\mathbf{S}(t, \mathbf{x}) = \frac{1}{\mu_0 c} \mathbf{E}(t, \mathbf{x}) \times (\hat{\mathbf{x}} \times \mathbf{E}(t, \mathbf{x})) = \frac{1}{\mu_0 c} [\hat{\mathbf{x}} |\mathbf{E}(t, \mathbf{x})|^2 - \mathbf{E}(t, \mathbf{x}) (\hat{\mathbf{x}} \cdot \mathbf{E}(t, \mathbf{x}))]. \tag{1.24}$$

Note that $\hat{\mathbf{x}} \cdot \mathbf{E} = 0$, and hence we have

$$\mathbf{S}(t, \mathbf{x}) = \frac{\mu_0}{16\pi^2 c |\hat{\mathbf{x}}|^2} \left| \int d^3y \dot{\mathbf{J}}(t_{\text{ret}}, \mathbf{y}) \right|^2 \sin^2 \theta \hat{\mathbf{x}}, \tag{1.25}$$

where θ is the angle between the integral and $\hat{\mathbf{x}}$.